Week 1: From Research Question to Statistical Model

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Skills Learning – Lecture

This week, we are shifting from basic R skills toward statistical thinking.

In the introductory course, we focused on importing data, exploring data, cleaning columns, summarizing variables, and making plots. In this advanced course, we will build on those skills and ask a new question:

> **Given my biological question and my data, what kind of statistical model should I use?**

Before we can choose a model, we need to understand:

- what kind of outcome variable we have
- whether observations are independent or repeated
- whether the dataset is cross-sectional or longitudinal
- what the distribution of the response variable looks like
- whether the predictor variables are continuous, categorical, count, binary, or proportions

This week is about building that foundation.

0. Load Required Packages

```
# Load packages
library(tidyverse)
library(janitor)
library(here)
```

1. Load and Preview the Dataset

Today we are using a faux gorilla hormone + behavioral dataset.

Each row represents one sample/observation from a gorilla. Some gorillas appear more than once, which means this dataset contains **repeated measures**.

The dataset includes:

- gorilla identity and group information
- sex and age class

- hormone values
- behavioral variables
- health-related variables
- ecological variables

```r
# check that you are in the correct working directory
getwd()
```

```
## [1] "/Users/rsteinitz/Documents/github/R Advanced Data Analysis/Week 1"
```

```r
# list files inside the Week 1 folder
list.files("Week 1")
```

```
## character(0)
```

```r
# Load dataset
data <- readr::read_csv(here::here("Week 1/gorilla_hormone_data.csv")) %>%
  janitor::clean_names()
```

```
## Rows: 240 Columns: 19
## -- Column specification ------------------------------------
## Delimiter: ","
## chr  (8): sample_id, gorilla_id, gorilla_name, group, sa...
## dbl (11): age_years, dominance_rank, group_size, fruit_a...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Preview the structure
glimpse(data, width = 80)
```

```
## Rows: 240
## Columns: 19
## $ sample_id                <chr> "G001_S01", "G001_S02", "G001_S03", "G001_S04~
## $ gorilla_id               <chr> "G001", "G001", "G001", "G001", "G001", "G001~
## $ gorilla_name             <chr> "Gasore", "Gasore", "Gasore", "Gasore", "Gaso~
## $ group                    <chr> "BWE", "BWE", "BWE", "BWE", "BWE", "BWE", "BW~
## $ sample_date              <chr> "5/23/25", "8/18/25", "9/10/25", "2/2/25", "5~
## $ observer                 <chr> "Mark", "Sarah", "Mary", "Sarah", "Mark", "Ro~
## $ sex                      <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", ~
## $ age_class                <chr> "adult", "adult", "adult", "adult", "adult", ~
## $ age_years                <dbl> 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.~
## $ dominance_rank           <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 6, 6, 6, 6, 6, 6, 6, ~
## $ group_size               <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 3, 3, 3, 3, 3, 3, 3, ~
## $ fruit_availability_index <dbl> 0.4040977, 0.3301009, 0.3134359, 0.7098761, 0~
## $ feeding_time_prop        <dbl> 0.2554745, 0.6272515, 0.5106832, 0.3779914, 0~
## $ aggression_events        <dbl> 2, 0, 1, 0, 0, 1, 1, 0, 3, 0, 1, 0, 0, 1, 0, ~
## $ parasite_load            <dbl> 28, 3, 12, 19, 15, 7, 38, 26, 6, 17, 10, 4, 5~
## $ respiratory_signs        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ injury_status            <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
## $ cortisol_ng_ml           <dbl> 19.158612, 22.606257, 28.816896, 9.639142, 10~
## $ testosterone_ng_ml       <dbl> 5.717721, 3.666248, 3.084256, 4.783930, 3.218~
```

```r
# look at a particular var
data %>%
  count(age_class)
```

```
## # A tibble: 3 x 2
##   age_class     n
##   <chr>     <int>
## 1 adult       216
## 2 infant        8
## 3 subadult     16
```

2. Numeric Variables

A major goal this week is to identify different variable types.
Different variable types often require different statistical models.

2.1 Continuous Variables

A **continuous variable** is numeric and can take many possible values.

Examples in this dataset include:

- `cortisol_ng_ml`
- `testosterone_ng_ml`
- `age_years`
- `fruit_availability_index`

```r
# ================================================================================
# Q: Which variables are continuous???
# ================================================================================
str(data)
```

```
## spc_tbl_ [240 x 19] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ sample_id               : chr [1:240] "G001_S01" "G001_S02" "G001_S03" "G001_S04" ...
##  $ gorilla_id              : chr [1:240] "G001" "G001" "G001" "G001" ...
##  $ gorilla_name            : chr [1:240] "Gasore" "Gasore" "Gasore" "Gasore" ...
##  $ group                   : chr [1:240] "BWE" "BWE" "BWE" "BWE" ...
##  $ sample_date             : chr [1:240] "5/23/25" "8/18/25" "9/10/25" "2/2/25" ...
##  $ observer                : chr [1:240] "Mark" "Sarah" "Mary" "Sarah" ...
##  $ sex                     : chr [1:240] "F" "F" "F" "F" ...
##  $ age_class               : chr [1:240] "adult" "adult" "adult" "adult" ...
##  $ age_years               : num [1:240] 14.2 14.2 14.2 14.2 14.2 14.2 14.2 14.2 32.5 32.5 ...
##  $ dominance_rank          : num [1:240] 8 8 8 8 8 8 8 8 6 6 ...
##  $ group_size              : num [1:240] 1 1 1 1 1 1 1 1 3 3 ...
##  $ fruit_availability_index: num [1:240] 0.404 0.33 0.313 0.71 0.433 ...
##  $ feeding_time_prop       : num [1:240] 0.255 0.627 0.511 0.378 0.414 ...
##  $ aggression_events       : num [1:240] 2 0 1 0 0 1 1 0 3 0 ...
##  $ parasite_load           : num [1:240] 28 3 12 19 15 7 38 26 6 17 ...
##  $ respiratory_signs       : num [1:240] 0 0 0 0 0 0 0 0 0 0 ...
##  $ injury_status           : num [1:240] 0 0 0 0 0 0 0 0 1 0 ...
##  $ cortisol_ng_ml          : num [1:240] 19.16 22.61 28.82 9.64 10.81 ...
```

```
##  $ testosterone_ng_ml      : num [1:240] 5.72 3.67 3.08 4.78 3.22 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   sample_id = col_character(),
##   ..   gorilla_id = col_character(),
##   ..   gorilla_name = col_character(),
##   ..   group = col_character(),
##   ..   sample_date = col_character(),
##   ..   observer = col_character(),
##   ..   sex = col_character(),
##   ..   age_class = col_character(),
##   ..   age_years = col_double(),
##   ..   dominance_rank = col_double(),
##   ..   group_size = col_double(),
##   ..   fruit_availability_index = col_double(),
##   ..   feeding_time_prop = col_double(),
##   ..   aggression_events = col_double(),
##   ..   parasite_load = col_double(),
##   ..   respiratory_signs = col_double(),
##   ..   injury_status = col_double(),
##   ..   cortisol_ng_ml = col_double(),
##   ..   testosterone_ng_ml = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```
glimpse(data)
```

```
## Rows: 240
## Columns: 19
## $ sample_id                <chr> "G001_S01", "G001_S02", "~
## $ gorilla_id               <chr> "G001", "G001", "G001", "~
## $ gorilla_name             <chr> "Gasore", "Gasore", "Gaso~
## $ group                    <chr> "BWE", "BWE", "BWE", "BWE~
## $ sample_date              <chr> "5/23/25", "8/18/25", "9/~
## $ observer                 <chr> "Mark", "Sarah", "Mary", ~
## $ sex                      <chr> "F", "F", "F", "F", "F", ~
## $ age_class                <chr> "adult", "adult", "adult"~
## $ age_years                <dbl> 14.2, 14.2, 14.2, 14.2, 1~
## $ dominance_rank           <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 6~
## $ group_size               <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 3~
## $ fruit_availability_index <dbl> 0.4040977, 0.3301009, 0.3~
## $ feeding_time_prop        <dbl> 0.2554745, 0.6272515, 0.5~
## $ aggression_events        <dbl> 2, 0, 1, 0, 0, 1, 1, 0, 3~
## $ parasite_load            <dbl> 28, 3, 12, 19, 15, 7, 38,~
## $ respiratory_signs        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ injury_status            <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1~
## $ cortisol_ng_ml           <dbl> 19.158612, 22.606257, 28.~
## $ testosterone_ng_ml       <dbl> 5.717721, 3.666248, 3.084~
```

```
# Let's look at some variables
summary(data$age_years)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    3.40   16.10   25.75   24.22   32.60   37.70
```

```r
summary(data$sex)
```

```
##    Length     Class      Mode
##       240 character character
```

```r
summary(data$parasite_load)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    9.00   16.00   19.44   26.00  111.00
```

```r
# ============================================================================
# Q: Which of these variables are continuous?
# ============================================================================

# Check class
class(data$cortisol_ng_ml) # this is a numeric variable
```

```
## [1] "numeric"
```

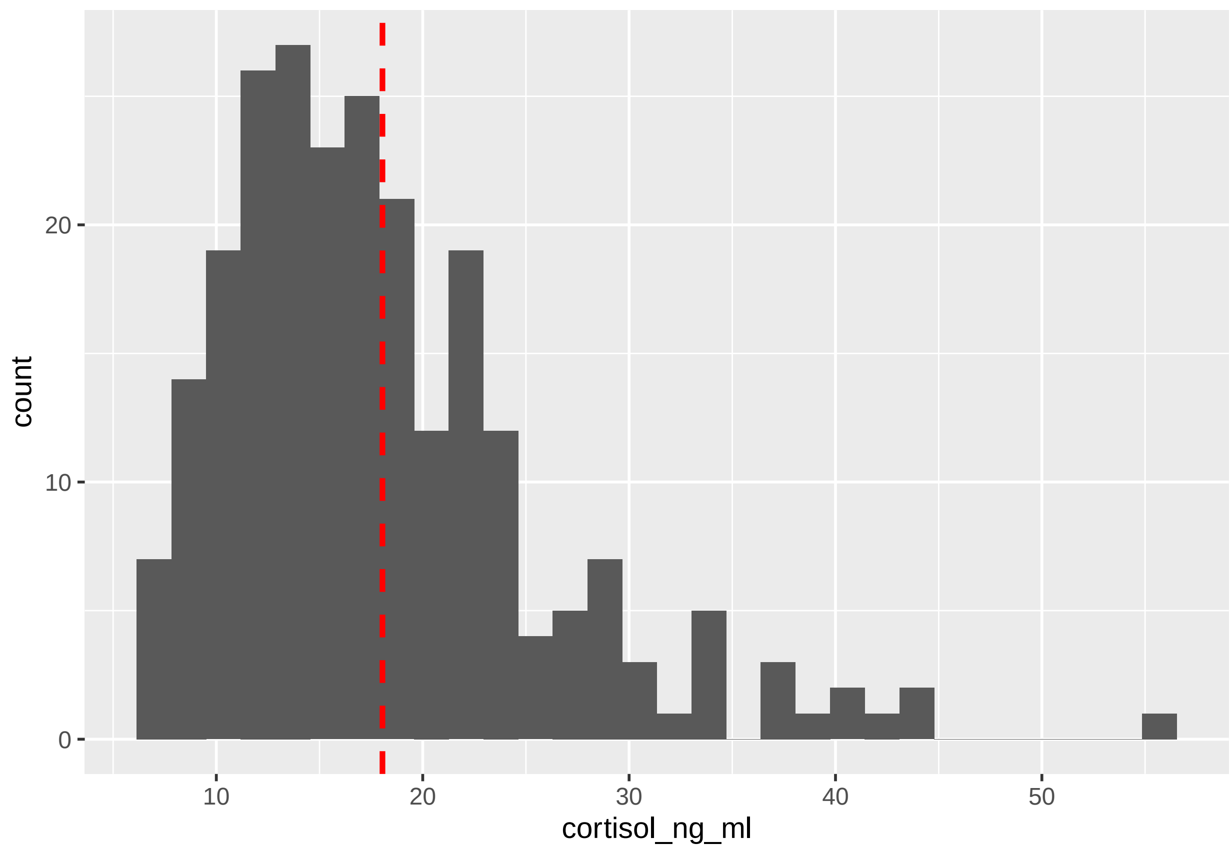
```r
class(data$age_years) # numeric
```

```
## [1] "numeric"
```

```r
# ============================================================================
# Q: What is the range of cortisol values?
# Q: What is the mean testosterone value?
# ============================================================================

ggplot(data, aes(x = cortisol_ng_ml)) +
  geom_histogram() +
  geom_vline(aes(xintercept = mean(data$cortisol_ng_ml)),
             color = "red",
             size = 1,
             linetype = "dashed")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2
## 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where
## this warning was generated.
```

```
## `stat_bin()` using `bins = 30`. Pick better value
## `binwidth`.
```



2.2 Count Variables

A **count variable** is numeric, but it represents the number of times something happened.

Count variables usually:

- are whole numbers
- cannot be negative
- often have many zeros
- are often not normally distributed

Examples in this dataset include:

- `aggression_events`
- `parasite_load`

```
summary(data$aggression_events)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   0.000   1.000   1.688   2.000   8.000
```

```
class(data$aggression_events)
```

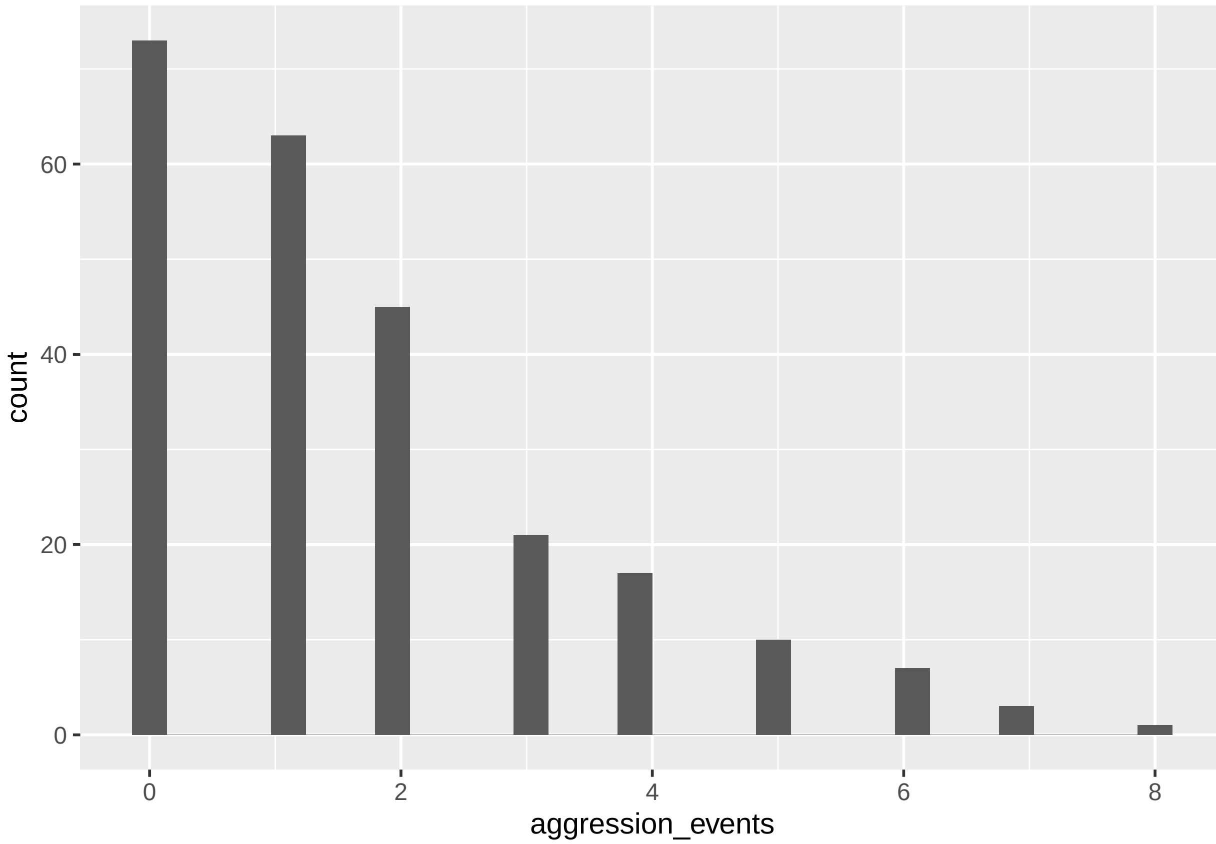
```
## [1] "numeric"
```

```r
# class is numeric, and these are continuous,
# but they are counts and tend to be 0-inflated

# note how the min and max are round numbers- 0 and 8
# the mean is a number with a decimal, because a mean of counts is not
# necessarily a round number. For example, the mean of (1, 4, and 5) is 3.333

# Count how many rows have each aggression value
count(data, aggression_events)
```

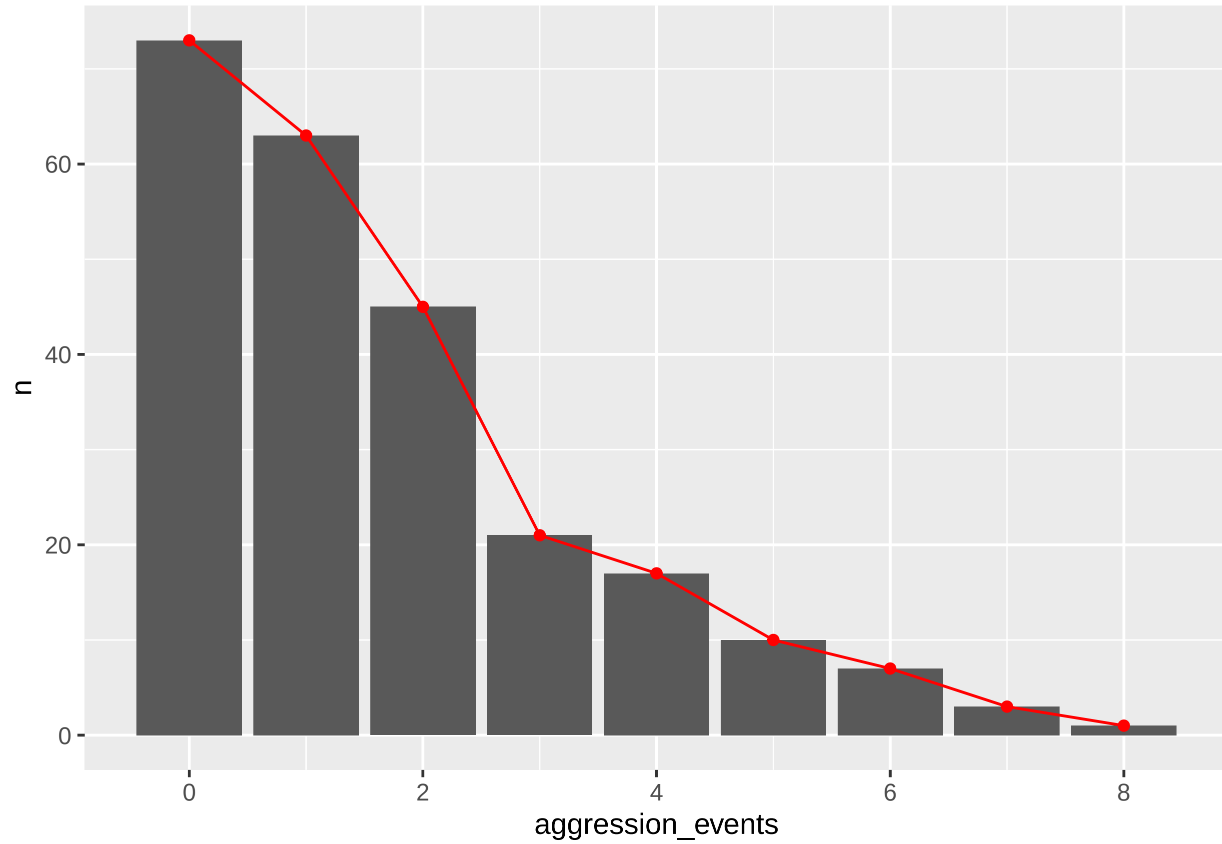
```
## # A tibble: 9 x 2
##   aggression_events     n
##               <dbl> <int>
## 1                 0    73
## 2                 1    63
## 3                 2    45
## 4                 3    21
## 5                 4    17
## 6                 5    10
## 7                 6     7
## 8                 7     3
## 9                 8     1
```

```r
# What does each row mean?

# ================================================================================
# Q: Why is aggression_events a count variable rather than a continuous variable?
# ---> Mathematically, a count variable is numeric, but it is not continuous
# because it cannot take every value within a range.

# for example, cortisol can take any of the following:
# 14.2
# 14.21
# 14.213
# 14.2137
# There are infinitely many possible values between 14 and 15.

# But you can't log 3.54 aggression events. Each aggression event either
# happens or it does not, so each event is a whole number-- 1; when you
# count them together, they form round numbers: 1, 2, 3... 246.. and so on.

# Count variables have different statistical properties that may violate
# assumptions of ordinary linear models, for example.
# ================================================================================

# ================================================================================
# Q: Can a count variable have decimal values?
# Q: Can a count variable be negative?
# ================================================================================

ggplot(data, aes(x = aggression_events)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value
## `binwidth`.
```




```
# ==============================================================================
# Q: Why does the distribution have breaks between columns?
# ==============================================================================

# Another cool option:
# manually calculate the height of each bar of the histogram...
hist_data <- data %>%
  count(aggression_events)

# then use that number as the max to show a line connecting the bars
ggplot(hist_data, aes(x = aggression_events, y = n)) +
  geom_col() +
  geom_line(color = "red") +
  geom_point(color = "red")
```



2.3 Binary Variables

A **binary variable** has two possible values: 0 or 1.

Examples in this dataset include:

- `respiratory_signs`
- `injury_status`

In this dataset, `respiratory_signs` and `injury_status` are coded as:

- `0` = no
- `1` = yes

```
class(data$respiratory_signs)
```

```
## [1] "numeric"
```

```
class(data$injury_status)
```

```
## [1] "numeric"
```

```r
data %>%
  count(respiratory_signs)
```

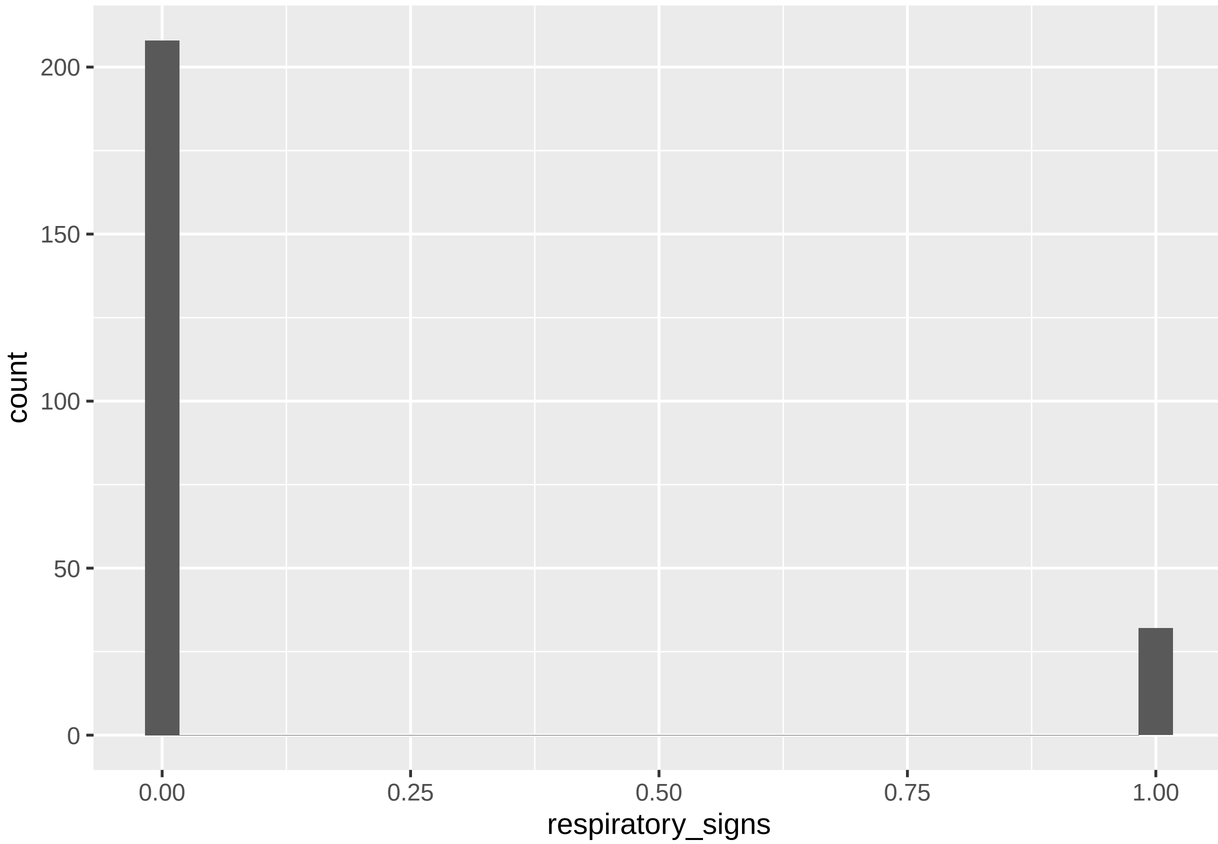
```
## # A tibble: 2 x 2
##   respiratory_signs     n
##               <dbl> <int>
## 1                 0   208
## 2                 1    32
```

```r
data %>%
  count(injury_status)
```

```
## # A tibble: 2 x 2
##   injury_status     n
##           <dbl> <int>
## 1             0   220
## 2             1    20
```

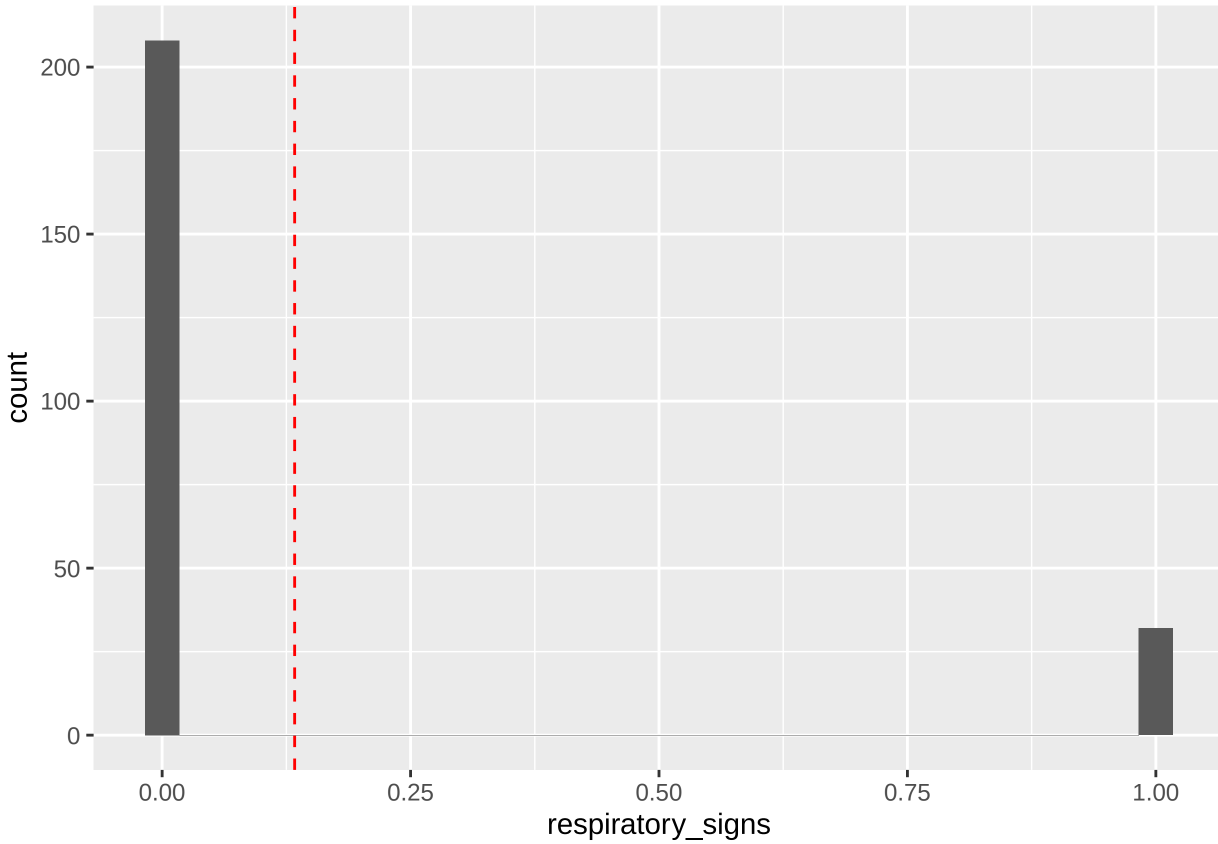
```r
# ===============================================================================
# Q: What proportion of samples have respiratory signs?
# Q: What proportion of samples have injury signs?
# ===============================================================================

ggplot(data, aes(respiratory_signs)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value
## `binwidth`.
```




```
ggplot(data, aes(respiratory_signs)) +
  geom_histogram() +
   geom_vline(aes(xintercept = mean(data$respiratory_signs)),
             color = "red",
             linetype = "dashed")
```

```
## ‘stat_bin()‘ using ‘bins = 30‘. Pick better value
## ‘binwidth‘.
```




```
# ==================================================================================
# Q: Why does the distribution have breaks between columns?
# Q: Why is the mean not exactly in the middle between the two columns?
# ==================================================================================

# Proportion of samples with respiratory signs
count(data, respiratory_signs)
```

```
## # A tibble: 2 x 2
##   respiratory_signs     n
##               <dbl> <int>
## 1                 0   208
## 2                 1    32
```

```
32/240 # number of observations with respiratory signs out of all observations
```

```
## [1] 0.1333333
```

```
# another way to do this:
mean(data$respiratory_signs)
```

```
## [1] 0.1333333
```

```r
# ==================================================================================
# Q: what is the proportion of observations of indivs with injuries?
# ==================================================================================
# Proportion of samples with injury signs
mean(data$injury_status)
```

```
## [1] 0.08333333
```

2.4 Proportion Variables

A **proportion variable** is numeric but bounded between 0 and 1.

In this dataset, `feeding_time_prop` represents the proportion of observed time spent feeding.

```r
class(data$feeding_time_prop)
```

```
## [1] "numeric"
```

```r
summary(data$feeding_time_prop)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2032  0.3938  0.5024  0.5151  0.6219  0.9229
```

```r
# ==================================================================================
# Q: Why is feeding_time_prop not a count variable?
# Q: What are the lowest and highest feeding time proportions?
# Q: Why might proportions require special attention in statistical models?
# ==================================================================================
```

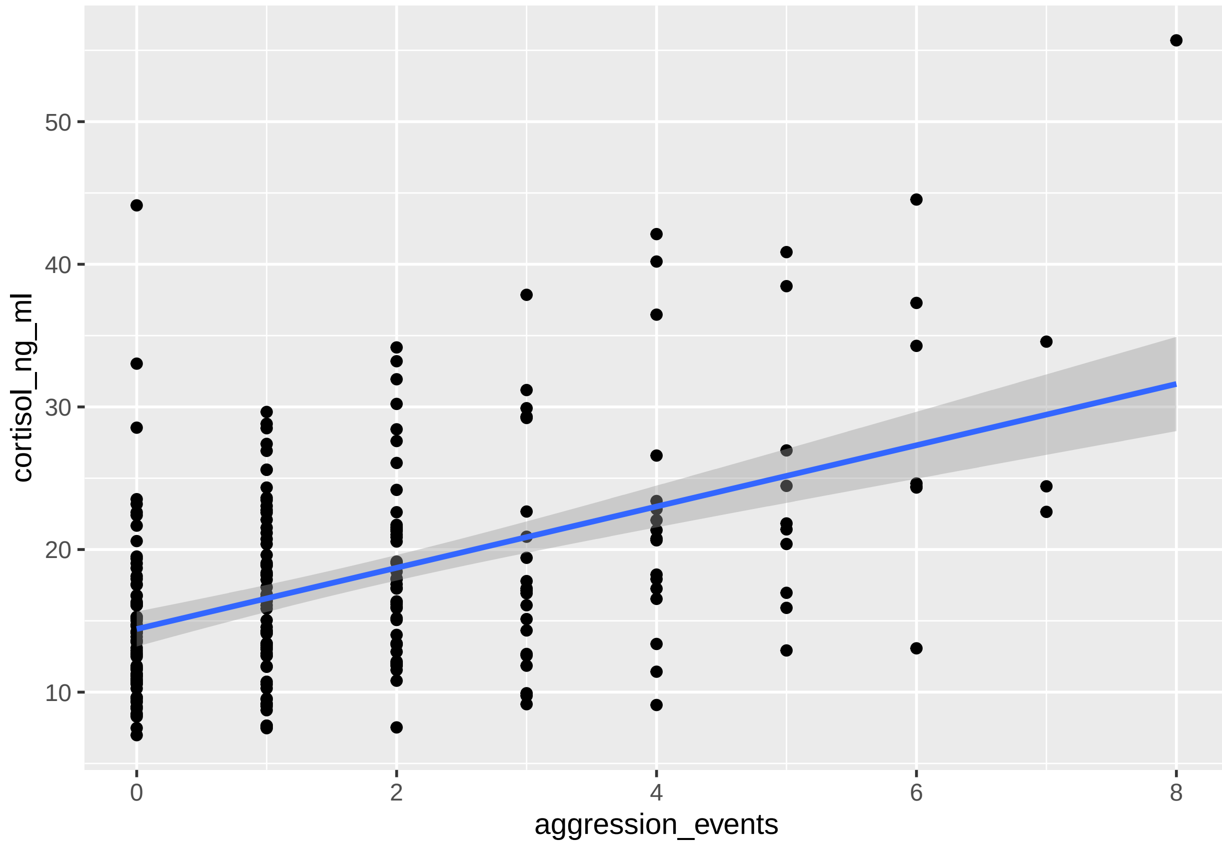
Proportions are **continuous**, but they are bound between 0 and 1, which makes them behave differently from unconstrained continuous variables like `age`, `cortisol`, or `body mass`.

3. ANALYSIS

Let's ask this question: Do gorillas with more aggression events have higher cortisol?

```r
# How might we try to answer the question:
# **Do gorillas with more aggression events have higher cortisol?**

ggplot(data, aes(x = aggression_events, y = cortisol_ng_ml)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# so we can write a model to test this
mod1 <- lm(cortisol_ng_ml~aggression_events, data = data)
summary(mod1)
```

```
##
## Call:
## lm(formula = cortisol_ng_ml ~ aggression_events, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -14.238  -4.772  -1.237   3.059  29.706
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        14.4252     0.6219  23.196  < 2e-16 ***
## aggression_events   2.1478     0.2559   8.392 4.29e-15 ***
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.932 on 238 degrees of freedom
## Multiple R-squared:  0.2283, Adjusted R-squared:  0.2251
## F-statistic: 70.42 on 1 and 238 DF,  p-value: 4.289e-15
```

```r
# --> For every additional aggression event, cortisol increases by
# about 2.15 ng/mL {the `estimate` of `aggression_events`}

# ==============================================

# ...now, what if we want to ask:
# **Does aggression predict cortisol after accounting for injury status?**

# here, we include a **CONTROL VARIABLE** injury status:
mod2 <- lm(cortisol_ng_ml~aggression_events + injury_status, data = data)
summary(mod2)
```

```
##
## Call:
## lm(formula = cortisol_ng_ml ~ aggression_events + injury_status,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.7062  -4.4827  -0.9608   3.0015  29.7501
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        14.3812     0.6040  23.809  < 2e-16 ***
## aggression_events   1.8561     0.2594   7.154 1.04e-11 ***
## injury_status       6.4346     1.6411   3.921 0.000115 ***
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.732 on 237 degrees of freedom
## Multiple R-squared:  0.2753, Adjusted R-squared:  0.2692
## F-statistic: 45.02 on 2 and 237 DF,  p-value: < 2.2e-16
```

```r
# cortisol = 18.40 + (1.86 × aggression_events) + (6.43 × injury_status)
```

This means: Holding injury status constant, each additional aggression event is associated with an increase of about 1.86 ng/mL cortisol.

the coefficient of `aggression events` changed:
`2.15` –> `1.86`
This tells us that some of the relationship between aggression and cortisol was actually shared with injury status.

In other words: > more aggressive individuals may be somewhat more likely to be injured; and injured individuals have higher cortisol

- uninjured gorilla → add 0 x 6.43 = 0

- injured gorilla → add 1 x 6.43 = 6.43
  So, after accounting for aggression, injured gorillas have cortisol values that are approximately 6.43 ng/mL higher than uninjured gorillas.

So when injury enters the model, aggression itself loses a little explanatory power . . . but `p = 1.0e-11`, so **aggression** remains strongly significant.

Now, the coefficient of injury_status = `6.43`
. . . and since injury_status is either 0 or 1, then injured gorillas have cortisol values that are, on average, about 6.4 ng/mL higher than non-injured gorillas, after accounting for aggression, which is a pretty substantial effect.

But this doesn't necessarily mean that injury is a better predictor. Because the predictors (aggression vs. injury) are measured on entirely different scales.

- 1 unit of aggression means *+1 aggression event*.
- 1 unit of injury means going from *uninjured* to *injured* (all or none- 0 or 100%).

So, what *CAN* we say??

> "Injury status contributes important additional information beyond aggression."

R^2 increased: 0.228 –> 0.275, so the model now explains 27.5% instead of 22.8% of the variation we see.

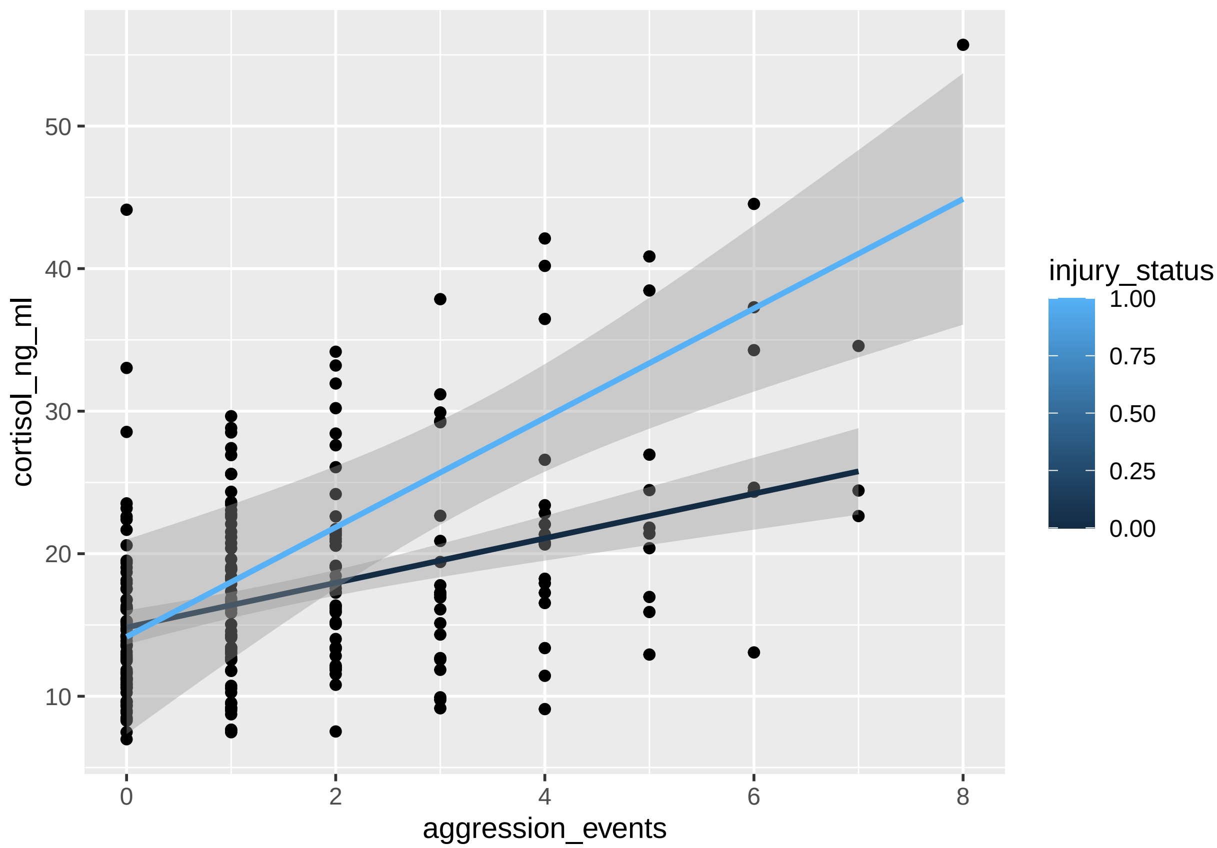
- In **Model 1**, `aggression` predicts `cortisol`.
- In **Model 2**, `aggression` *still* predicts `cortisol`, even after accounting for `injury status`. `Injury status` also predicts `cortisol`, even after accounting for `aggression`.

  The coefficient for each predictor tells us its relationship with the outcome after accounting for all of the other predictors in the model.

That is the essence of multiple regression.

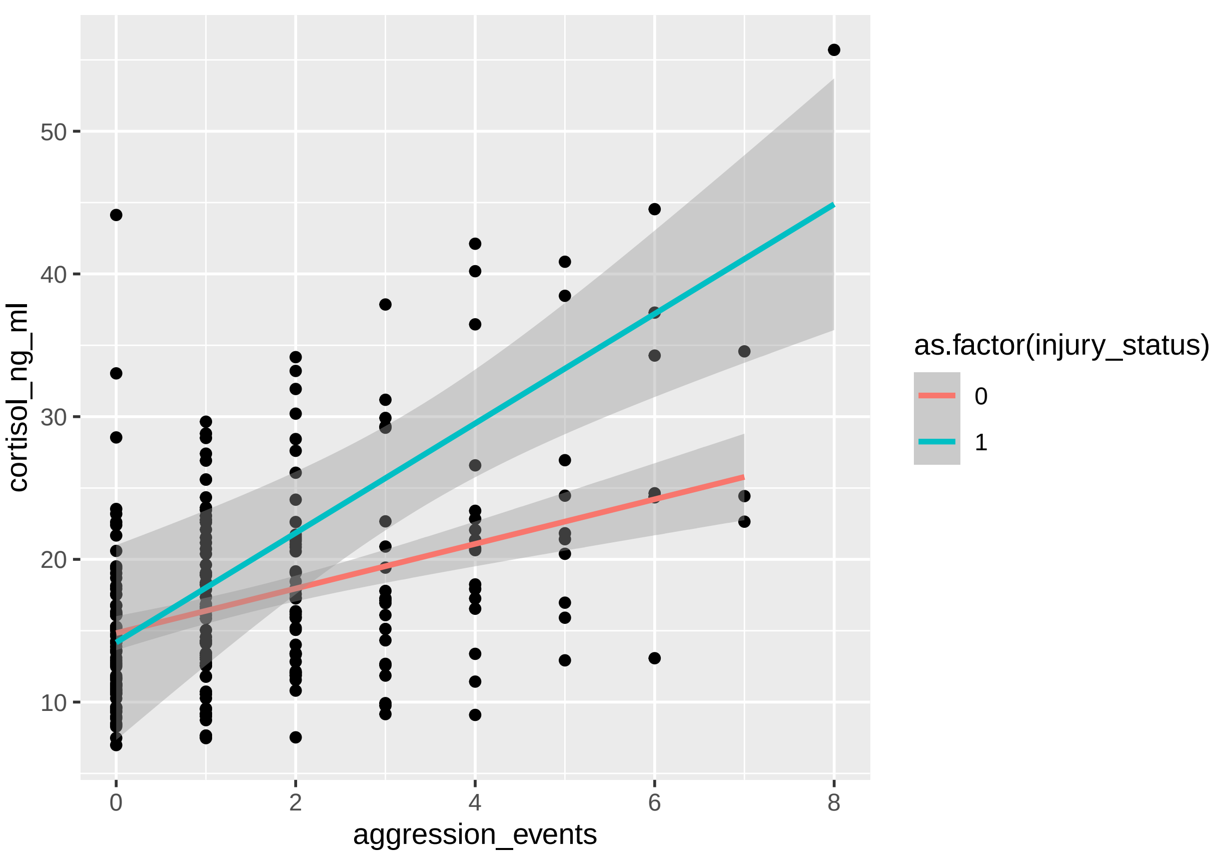
```r
# and we can plot this as well, where this relationship becomes more apparent:
ggplot(data, aes(x = aggression_events, y = cortisol_ng_ml, group = injury_status)) +
  geom_point() +
  geom_smooth(method = "lm", aes(color = injury_status))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# if we used: `aes(color = injury_status)`, without "as.factor",
# ggplot assumes `injury_status` is a continuous variable.
# which makes sense for something like `body_mass` or `age`.

ggplot(data, aes(x = aggression_events, y = cortisol_ng_ml, group = injury_status)) +
  geom_point() +
  geom_smooth(method = "lm", aes(color = as.factor(injury_status)))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# but `injury_status` is not-- it can either be 0 or 1.
# so we're telling ggplot: Don't treat these as numbers. Treat them as categories.
```

4. Considerations

4.1 Important model-selection questions

Before choosing a model, ask:

1. What is the response variable?
2. What type of variable is the response?
3. Is the response approximately normal, skewed, count, binary, or proportion?
4. Are observations independent?
5. Are individuals sampled repeatedly?
6. Is the dataset cross-sectional or longitudinal?
7. What are the predictor variables?
8. Are predictors continuous, categorical, binary, count, or proportion?
9. Does the research question involve group differences, relationships, prediction, or repeated measures?

Skills Application – Lab

5. Lab Challenge: Choose the Model Type

For this lab, your goal is **not** to run the final model yet.

Your goal is to look at a biological question, identify the response variable, diagnose the structure of the data, and decide what kind of model might be appropriate.

For each question below, answer:

1. What is the response variable?
2. What type of variable is the response?
3. What are the predictor variables?
4. Are the observations independent?
5. Is the dataset cross-sectional or longitudinal/repeated?
6. What plot would help you inspect the response?
7. What general model type might be appropriate?

5.1 Question 1

> Do gorillas with more aggression events have higher cortisol?

```
# Make at least one plot to inspect this question here
```

Your answer:
Response variable:
Response type:
Predictor variable(s):
Independent or repeated observations?
Cross-sectional or longitudinal?
Useful plot:
Possible model type:

5.2 Question 2

> Are respiratory signs more likely in larger groups?

```
# Make at least one plot to inspect this question here
```

Your answer:
Response variable:
Response type:
Predictor variable(s):
Independent or repeated observations?
Cross-sectional or longitudinal?
Useful plot:
Possible model type:

5.3 Question 3

> Does feeding time proportion differ by group?

```
# Make at least one plot to inspect this question here
```

Your answer:
Response variable:
Response type:
Predictor variable(s):
Independent or repeated observations?
Cross-sectional or longitudinal?
Useful plot:
Possible model type:

5.4 Optional: Your own biological question

Write one biological question using this dataset or your own dataset.

Your question:

```
# Make at least one plot to inspect your question here
```

Your answer:
Response variable:
Response type:
Predictor variable(s):
Independent or repeated observations?
Cross-sectional or longitudinal?
Useful plot:
Possible model type:
